

POKHARA UNIVERSITY

Level: Bachelor

Semester – Fall

Year : 2012

Programme: BE

Full Marks: 100

Course: Artificial Intelligence

Pass Marks: 45

Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

- a) Explain Turing test. Machine can be made intelligent artificially but ultimately human make the machines. So who is more intelligent – the artificial machine or the person? Discuss. 4+4
- b) In a crypto-arithmetic puzzle, the variable A, B, C, D, E and F can take values from 1 to 7. The variables must all be different and, when taken as digits, they must satisfy the following sum. Solve the following problem as CSP. 7

$$\begin{array}{r} AB \\ + CD \\ \hline EF \end{array}$$

2. a) Explain production systems with the help of 8-puzzle example. 8
- b) What are the problems that may arise in hill climbing searching? How they can be handled? Explain. 4+3

OR

- a) You are given two jugs, a 4 liters gallon one and a 3 liters gallon one. Neither have any measuring markers on it. There is a pump that can be used to fill the jugs with water. How can you get exactly 2 liters of water into the 4 liters gallon jug? Solve this problem with state space representation. 8
- b) Explain Min-Max search strategy with example. 7
3. a) What is meaning of word *heuristics* in the context of search strategies? What conditions on A* search is required to guarantee completeness and optimality. 3+4
- b) What are the different dimensions of search strategy? Compare the 4+4

BFS and DFS with different dimension.

- 4 a) The logical operator " $\leftrightarrow$ " is read "if and only if". $P \leftrightarrow Q$ is defined as being equivalent to $(P \rightarrow Q) \wedge (Q \rightarrow P)$. Based on this definition, show that $P \leftrightarrow Q$ is logically equivalent to $(P \vee Q) \rightarrow (P \wedge Q)$ using truth table.
- b) Represent the following using first order Logic.
- Prasant likes easy courses.
 - Science courses are hard.
 - All the courses in the computer departments are easy.
 - AI is a computer course.
- Also, use **resolution** to answer the question, "What course would Prasant like?"

OR

- a) Prove that "jeevan is happy" with the help of following facts expressed in CNF.
- $\neg \text{pass}(X, \text{history}) \vee \neg \text{win}(X, \text{lottery}) \vee \text{happy}(X)$
 - $\neg \text{study}(X) \vee \text{pass}(Y, Z)$
 $\neg \text{lucky}(W) \vee \text{pass}(W, U)$
 - $\neg \text{study}(\text{jeevan})$
 $\text{lucky}(\text{jeevan})$
 - $\neg \text{lucky}(U) \vee \text{win}(U, \text{lottery})$
- b) Explain with Example, how first order logic sentences are converted into conjunctive normal form (CNF).
5. a) How can you say FOPL extend the expressiveness of PL? 5
- b) Explain the forward and backward representational mapping with diagram. 5
- c) Explain Semantic Network with suitable example. 5
6. a) Explain the basic components of expert system. How can we make expert system knowledge base reusable? 4
- b) What do you mean by learning by analogy? Explain derivational analogy. 3+
7. Write short notes on **any two**: 2+
- Bayesian Network
 - Frames

